

Lab: Social Cognition, Language, and Intersubjectivity

Learning Goal: Apply Active Inference to philosophical problems of communication, understanding, and social ontology.

Part 1: The Other Minds Problem

Exercise: Consider three approaches to understanding another person's pain:

Approach	How It Works	Strengths	Limitations
Argument from analogy	I feel pain when hurt, others behave similarly when hurt, so they probably feel pain	Simple, intuitive	Only works by analogy to my one case; can't verify
Simulation theory	I simulate what I would feel in their situation	Captures empathy, embodied	My simulation may not match their actual state
Theory theory	I apply a folk-psychological theory: "injury causes pain"	Generalizable, theory-driven	Theories can be wrong; feels too cold
Active Inference	I have a generative model of pain-behavior that predicts their behavior given inferred hidden states	Formal, testable, hierarchical	Doesn't explain <i>why</i> there's something it's like to be in pain

Write a 300-word analysis: Which approach does Active Inference most closely resemble? Which philosophical problems does it solve? Which remain?

Write your response here

Part 2: Language as Model Alignment

Learning Goal: Analyze a real conversation through the Active Inference lens.

Exercise: Transcribe a brief (5-10 line) real conversation (or use this example):

A: "Have you eaten?"

B: "I had a sandwich earlier."

A: "Want to grab dinner?"

B: "Sure, where?"

A: "How about that Thai place?"

B: "Perfect."

For each utterance, analyze:

1. What is the speaker trying to reduce uncertainty about?
2. What shared priors make the utterance interpretable?
3. Where does pragmatic value (achieving goals) vs. epistemic value (reducing uncertainty) dominate?
4. What Gricean maxims are being observed?

Write your response here

Part 3: Institutional Reality

Learning Goal: Apply Active Inference to social ontology — analyzing how institutions exist.

Exercise: Choose one social institution (money, marriage, law, university, democracy).

Analyze it as a shared generative model:

1. **What are the shared priors?** (What do participants need to believe for the institution to function?)
2. **What observations confirm the institution?** (What sensory evidence indicates it's "real"?)
3. **What would constitute a "prediction error"** for the institution? (When does the institution fail or break down?)
4. **How is the institution maintained?** (What communicative actions sustain the shared priors?)
5. **Could the institution have a Markov Blanket?** (What are its sensory inputs, active outputs, internal states?)

Write a 400-word analysis treating the institution as an Active Inference agent at a social scale.

Write your response here

Part 4: Hermeneutic Analysis

Learning Goal: Apply the hermeneutic circle and Gadamer's concepts to interpretation.

Exercise: Choose a difficult-to-interpret text, image, or cultural artifact. Analyze the interpretation process:

1. What **pre-understanding** (priors) do you bring to the interpretation?
2. How does the **hermeneutic circle** operate? (How does understanding parts change your understanding of the whole, and vice versa?)
3. Where does **fusion of horizons** occur? (Where does your interpretive horizon meet the artifact's?)
4. How could two people with different priors interpret the same artifact differently?

5. How does this relate to Active Inference's hierarchical model updating?

Write your response here

Part 5: Communication Failure Analysis

Learning Goal: Analyze breakdowns in communication using Active Inference.

Exercise: Describe a real or hypothetical communication failure (misunderstanding, cross-cultural confusion, ideological disagreement). Analyze it in Active Inference terms:

1. Where did the shared priors diverge?
2. What prediction errors went undetected?
3. How could precision weighting have contributed (e.g., one party weighting their priors too heavily)?
4. What would "repair" look like in formal terms (restoring shared priors)?

Write your response here

Part 6: Reflection

In 300 words, reflect: Active Inference models communication as mutual model alignment. But human communication also involves power dynamics, emotional manipulation, aesthetic expression, and play. Can Active Inference account for these dimensions, or are they beyond its scope? What would a more complete account of communication require?

Write your response here

Lab Summary

Part	Skill Practiced	Key Concept
1	Comparative analysis	Other minds solutions
2	Applied analysis	Language as coupled inference
3	Social ontology	Institutional reality
4	Hermeneutic interpretation	Prior-dependent understanding
5	Failure analysis	Communication breakdown
6	Critical reflection	Limits of the framework